

Diwas Mainali

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PROFESSIONAL EXPERIENCE

Lead Full Stack Engineer (Founder)

May 2025 – Present

Cram and Conquer | www.cramandconquer.com

Sydney, Australia

- Developed a full-stack productivity platform using React, Node.js, and MongoDB, scaling to 600+ active users and 100+ premium subscribers through features like real-time group study and advanced analytics.
- Secured authentication and payment systems with Google OAuth integration and Stripe payments, including comprehensive rate limiting, DDoS protection, and XSS prevention for enterprise-level security.
- Architected a real-time social layer via WebSockets to support seamless friend requests and active user tracking, ensuring synchronized state management with minimal network latency.
- Optimised performance and scalability through efficient database design, automated session cleanup, token refresh mechanisms, and responsive design patterns, ensuring seamless user experience across all devices.

Fullstack Developer Intern

September 2025 – December 2025

Lorgan Pty Ltd

Sydney, Australia

- Built a real-time patient monitoring platform using Django with complete CRUD functionality.
- Managed client expectations through regular demos, ensuring features were delivered according to business goals.
- Designed the relational data model and wrote queries for the patient monitoring application using PostgreSQL.
- Created responsive React components to deliver a modern, seamless, and interactive user interface.

AI Frontend Development Trainer

April 2025 – Present

Outlier

Remote

- Trained AI systems to improve frontend development capabilities by creating precise implementation examples.
- Debugged and refined AI-generated code to match design specifications from reference screenshots.
- Produced optimised React and HTML/CSS solutions that serve as high-quality training examples for AI learning.
- Collaborated with AI engineers to identify and address patterns in frontend code generation deficiencies.

Retention Specialist

November 2024 – April 2025

Volkswagen Financial Services (via Forrest Contact)

Remote

- Managed 10-15 daily refinancing inquiries in Salesforce, providing quotes for residual loans at 98% accuracy.
- Processed loan restructuring for 100+ clients monthly, implementing flexible payment solutions in Salesforce.

TECHNICAL PROJECT

GridBase — Airtable Clone | [gridbase-diwas.vercel.app/](#)

Next.js, TypeScript, tRPC, PostgreSQL

- Built and optimised a high-performance virtualised data grid utilising TanStack Table, successfully rendering and managing datasets up to millions of rows without compromising scroll or load speed.
- Implemented a real-time, collaborative editing experience by integrating optimistic updates with a debounced, batch-saving system to efficiently synchronise user changes and minimise database load.
- Designed a dynamic query and filtering engine using raw PostgreSQL (SQL), enabling users to construct and execute complex, multi-layered filters and sorts against the persistent data store with sub-second latency.
- Architected a full-stack T3 application structure, establishing end-to-end type safety across the entire stack via tRPC and enforcing robust data integrity and validation through Zod schemas.

EDUCATION AND EXTRACURRICULAR

University of New South Wales

January 2022 – December 2025

Bachelor of Computer Science

Kensington, NSW, Australia

UNSW Chess Society Volunteer

- Organised and managed daily chess events for over 100 members.
- Created a web application for chess openings [linked here!](#), enhancing their skills and enjoyment of the game.

TECHNICAL PROFICIENCIES

Languages: TypeScript, Python, Java, C, HTML, CSS, Bash

Frameworks: React, Next.js, Node.js, Express.js, tRPC, Tailwind CSS

Developer Tools: PostgreSQL, MongoDB, Git, Cypress, JMeter, K6 Grafana